

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459345.6903 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.125 +/- 0.019
Transit depth (Rp/Rs)^2	1.57 +/- 0.48 [%]
Orbital Inclination (inc)	86.88 +/- 2.0
Airmass coefficient 1 (a1)	0.958 +/- 0.013
Airmass coefficient 2 (a2)	0.0246 +/- 0.0054
Scatter in the residuals of the lightcurve fit is	1.33 %
Best Comparison Star	#1 - [361, 76]
Optimal Aperture	3.44
Optimal Annulus	15.26
Transit Duration (day)	0.0577 +/- 0.0028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.125 ± 0.019 vs 0.1594 (deviation 1.81σ)
Duration fit vs published	0.0577 d vs 0.0512 d (deviation 2.3σ)
Mid-time O–C	1.8 min
Depth SNR	6.6
Residual scatter	1.33 %

Light curve

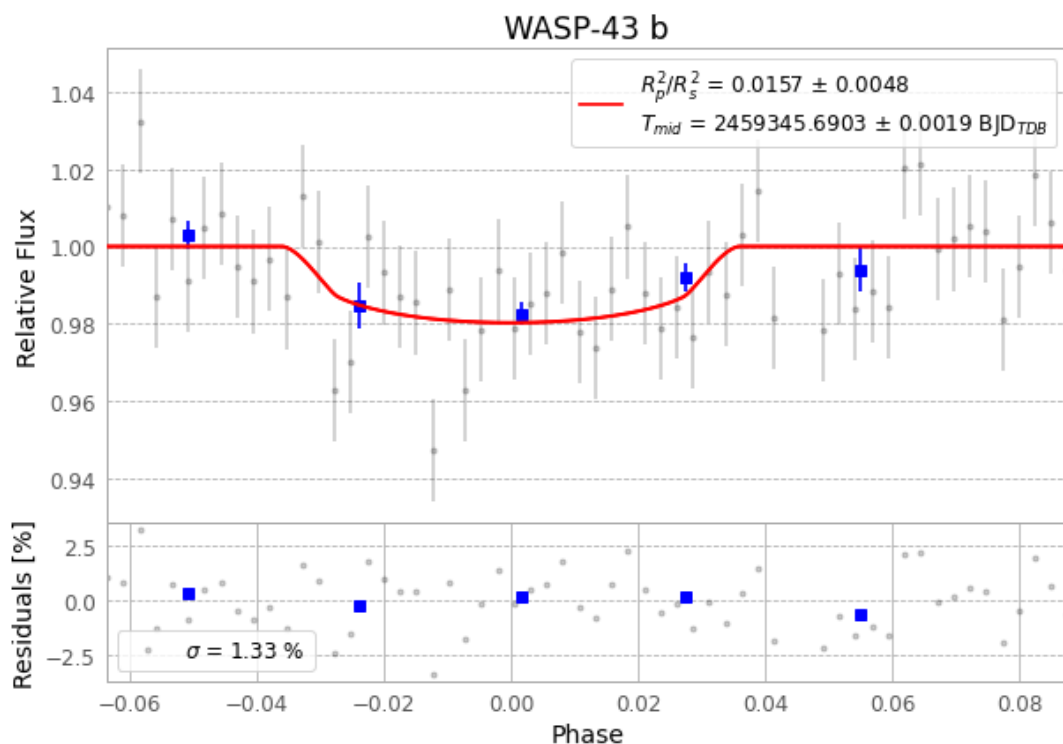


Figure 1 — FinalLightCurve_WASP-43 b_2021-05-10.png (SHA-256 0fd30318dddbccaa...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [428, 78], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6b3220f10d25aab7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

60 FITS frames (39 MB), WASP-43210511031512.FITS → WASP-43210511061212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-210511.log (067b4e276517...), FinalLightCurve_WASP-43 b_2021-05-10.png (0fd30318dddb...), FinalLightCurve_WASP-43 b_2021-05-10.csv (983af2edf551...)

Compute resources

Wall time 319.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-10 00:43Z.